



Group Assignment Cover Page

Course Code : **SECI1143**
Course Name : **PROBABILITY & STATISTICAL DATA ANALYSIS**
Section : 02
Lecturer's Name : Madam Rozilawati
Session/Semester : 20252026/2
Assignment Title : Assignment 1
Member Details : FARHAN ASHRAF BIN FAKHRUL ANWAR – A25CS0059

MUHAMMAD HAFIZUDDIN HAKIMI BIN HASMADI - A25CS0273

WAN FARUQ JAZLI BIN WAN AHMAD JAFFRY - A25CS0372

QUESTION 1 [15 MARKS]

A university library administration is conducting a study to understand the usage habits of its students. They randomly selected 60 students who visited the library during the first week of the semester. For each student, they recorded the following variables:

- Student ID
- Major of study (e.g., Engineering, Arts, Business)
- Academic Year (Year 1, Year 2, Year 3, Year 4)
- Total number of books borrowed this semester
- Time spent studying in the library per week (in hours)
- Temperature of the reading room when they visit (in Celsius)

Answer the following questions:

- a) Identify the **population** and the **sample** in this study. (2 marks)
- b) Classify each of the six variables as either **qualitative** or **quantitative**. (3 marks)
- c) For the quantitative variables identified in (b), state whether they are **discrete** or **continuous**. (2 marks)
- d) Determine the **level of measurement** (nominal, ordinal, interval, or ratio) for the following variables: *Major of study*, *Academic Year*, *Total number of books borrowed*, and *Temperature of the reading room*. (8 marks)

Question 1

a) Population : All university student who visited library
Sample : The randomly selected ~~student~~ 60 students who visited the library during the first week of the semester.

b) Student ID - Qualitative
Major of study - Qualitative
Academic year - Qualitative
Total number of books borrowed - Quantitative
Time spent studying in the library - Quantitative
Temperature of the reading room - Quantitative

c) Total number of books borrowed :- Discrete
Time spent studying in the library :- Continuous
Temperature of the reading room :- Continuous

No: _____

Date: _____

d)

Variables	Level of measurement
Major of study	Nominal
Academic year	Ordinal
Total number of books borrowed	Ratio
Temperature of the reading room	Interval

QUESTION 2 [15 MARKS]

A newly opened café wants to analyze the preferred beverages of its morning customers. The manager observes 120 customers and records their choices:

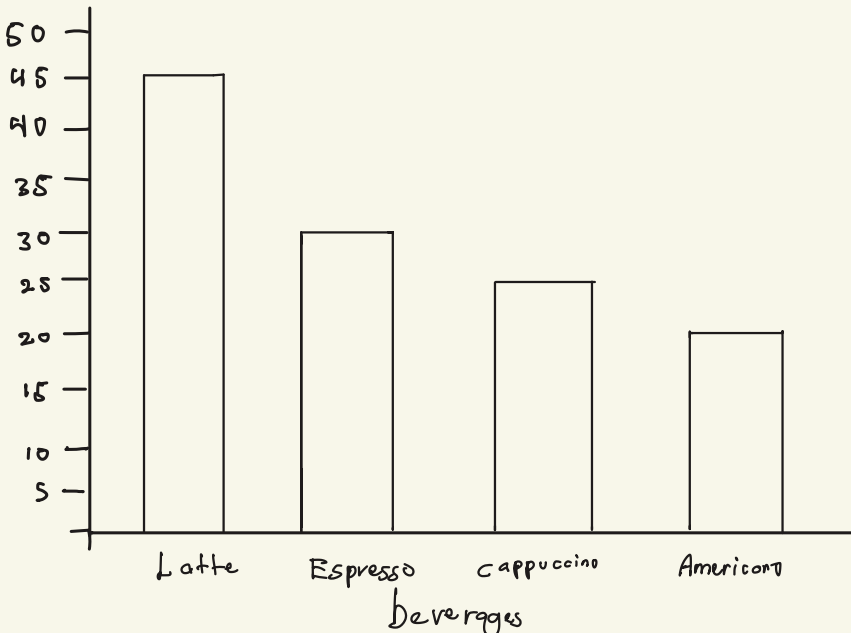
- Latte: 45 customers
- Espresso: 30 customers
- Cappuccino: 25 customers
- Americano: 20 customers

Answer the following questions:

- Construct a frequency table that includes the **frequency**, **relative frequency**, and **cumulative relative frequency**. (6 marks)
- Draw a fully labeled **bar chart** to visualize the frequencies of the preferred beverages. (5 marks)
- Construct a pie chart to display the relative frequencies of the beverages. Ensure each sector is labeled with the correct percentage. (4 marks)

beverages	frequency	relative frequency	cumulative relative frequency
Latte	45	0.375	0.375
Espresso	30	0.25	0.625
Cappuccino	25	0.208	0.833
Americano	20	0.167	1.00
Total	120	1.00	1.00

b) frequency



c.)

Beverage	Percentage of relative frequency
Latte	37.5
Espresso	25
Cappuccino	20.8
Americano	16.7
total	100

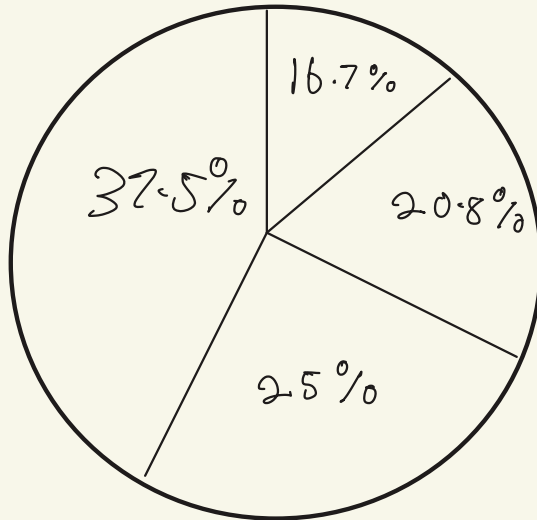
Relative frequency
of beverages

Latte = 37.5%

Espresso = 25%

Cappuccino = 20.8%

Americano = 16.7%



QUESTION 3 [15 MARKS]

A logistics company recorded the delivery time (in minutes) for 16 different packages dispatched within a city. The recorded times are as follows: 15, 18, 20, 22, 22, 25, 26, 28, 30, 32, 35, 40, 42, 65, 70, 75

Answer the following questions:

- Find the **mean**, **median**, and **mode** for the delivery times. Show your calculations. (4 marks)
- Calculate the **first quartile (Q1)**, **third quartile (Q3)**, and the **Interquartile Range (IQR)**. (4 marks)
- Using the **$1.5 \times \text{IQR}$ rule**, determine if there are any mild or extreme outliers in this dataset. Show your calculations for the lower and upper limits. (4 marks)
- Based on your findings, draw a **modified box plot** for the delivery times showing any outliers. (3 marks)

$$a) \text{ mean} = \frac{15 + 18 + 20 + 22 + 22 + 25 + 26 + 28 + 30 + 32 + 35 + 40 + 42 + 65 + 70 + 75}{16}$$

$$= 35.31$$

$$\text{median} = (28 + 30) \div 2 = 29$$

$$\text{mode} = 22$$

$$b) Q_1: i = \frac{25}{100}(16) = 4$$
$$(Y[4] + Y[5]) \div 2 \Rightarrow (22 + 22) \div 2 = 22$$

$$Q_3: i = \frac{75}{100}(16) = 12$$
$$(Y[12] + Y[13]) \div 2$$
$$\Downarrow$$
$$(40 + 42) \div 2 = 41$$

$$Q_3 - Q_1 = 41 - 22 = 19$$

$$c) 1.5 \times 19 = 28.5$$

$$\text{Lower} : Q_1 - 28.5 = 22 - 28.5 = -6.5$$

$$\text{Upper} : Q_3 + 28.5 = 41 + 28.5 = 69.5$$

$$\text{Extreme} : 3 \times \text{IQR} \Rightarrow 3 \times 19 = 57$$

$$\text{Lower} : Q_1 - 57 \Rightarrow 22 - 57 = -35$$

$$\text{Upper} : Q_3 + 57 \Rightarrow 41 + 57 = 98$$

$$69.5 < 98 \Rightarrow \text{mild outlier}$$

